Quantification of the co-mutagenic beta-carbolines, norharman and harman, in cigarette smoke condensates and cooked foods.

Co-mutagenic beta-carbolines, such as norharman and harman, were quantified in mainstream and sidestream smoke condensates of six Japanese brands of cigarettes, and also in 13 kinds of cooked foods, using a combination of blue cotton treatment and HPLC. Norharman and harman were detected in all the cigarette smoke condensate samples. Their levels in the mainstream smoke case were 900-4240 ng per cigarette for norharman, and 360-2240 ng for harman, and in sidestream smoke, 4130-8990 ng for norharman and 2100-3000 ng for harman. These beta-carbolines were also found to be present in all the cooked food samples, at levels of 2.39-795 ng for norharman and 0.62-377 ng for harman per gram of cooked food. The observed concentrations are much higher than those found for mutagenic and carcinogenic heterocyclic amines (HCAs), suggesting that humans are exposed to norharman and harman in daily life to a larger extent than to HCAs.